all at once - maybe?

A UNIQUANT research programme for six mathematical targets and testable hypotheses about fields, flow and observation

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Abstract

Can a common representation of time, phase, probability and coupled modes help with several mathematical problems at once? This paper formulates that question using the UNIQUANT system. It records exact rounding bounds, unequal phase classes, an explicit measurement nullspace and a weighted norm conservation law. It then states the separate bridges required for Navier–Stokes, Yang–Mills, P versus NP, the Riemann Hypothesis, Hodge and Birch–Swinnerton-Dyer. Elementary counterexamples show why finite norm conservation, spectral symmetry and coordinate changes cannot supply those bridges by themselves. A separate physical extension describes material-, charge-, orientation- and frequency-dependent coupling to proposed external fields, with conservation laws and an observation model that make the proposal testable. Shared motion and distance- dependent errors are treated explicitly. Reproducible synthetic calculations accompany the analysis. The result is a research programme with proved partial statements and specified missing steps, rather than a solution of the six targets.

Scope of "six". The six targets here are the Millennium problems other than Poincare. This identifies the research direction stated in the source UNIQUANT paper [1]; it does not suggest that one physical experiment can decide all six.

Dated status note. On 11 September 2026, the Clay Mathematics Institute said that Navier–Stokes had apparently been settled and described its evaluation process as ongoing. Its overview lists that problem as active and the other five treated here as unsolved [2, 3]. Consequently, this paper uses "six targets", not "six currently unsolved problems". It makes no claim about the validity or final recognition of the announced work.

Reading convention. Definitions specify the model. Propositions below are proved within their stated assumptions. Hypotheses are proposals requiring evidence. An explicit proof gap is a missing result, not a theorem established by terminology.

1 The mathematical core of UNIQUANT

The following definitions retain the rational time and phase structure of [1]. Here "quant" names a chosen coordinate unit; it is not an established minimum physical time or an electric charge.

1.1 Exact time with a retained remainder

Define a reference day and a rational time unit by

$$D_0 = 86400\ \text{s}, \qquad q = \frac{D_0}{768 \cdot 66 \cdot 23} = \frac{75}{1012}\ \text{s}. \tag{1}$$

For a model time t, measured relative to a fixed epoch in one declared time scale, set

$$\tau = t/q, \quad A = \left\lfloor \tau + \tfrac{1}{2} \right\rfloor, \quad \rho = \tau - A \in [-\tfrac{1}{2}, \tfrac{1}{2}), \qquad t = q(A + \rho). \tag{2}$$

The source epoch is 13 July 2029, 00:00 MESZ, or 12 July 2029, 22:00 UTC. Converting this civil label to the selected continuous calculation time scale is a separate step.

Proposition 1.1 (Address and interval error)**.** *If only qA is displayed, its error is at most $q/2$. For any two indices,*

$$|q(A_m - A_n) - (t_m - t_n)| = q|\rho_n - \rho_m| < q.$$

Keeping both A and ρ retains the original model time exactly.

Proof. Equation (2) gives $qA - t = -q\rho$. Subtract the two identities. The half-open interval of allowed remainders has diameter one. □

This controls representation error. It does not improve the accuracy of the clock or observation that supplied t. Exact rational arithmetic removes avoidable numerical rounding, while observational uncertainty remains.

1.2 A model-specific annual remainder

The source also defines

$$\tau(n) = Bn + \frac{an - 11n^2}{529}, \qquad B = 425806080, \quad a = 703993. \tag{3}$$

The centered fractional remainder has least integer period 529. Indeed the difference under $n \mapsto n + 529$ is integer. If P is any period, subtracting consecutive differences forces $529 \mid 22P$, hence $529 \mid P$. This is a property of the chosen rational polynomial; a physical 529-year oscillation does not follow from it.

1.3 Unequal phase classes

On $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, sort the distinct boundaries

$$\mathcal{B} = \{j/13 : 0 \le j \le 13\} \cup \{\ell/23 : 0 \le \ell \le 23\}. \tag{4}$$

The 36 boundaries define 35 intervals $I_j = [b_j, b_{j+1})$, of width w_j and midpoint m_j. Their widths are $r/299$ twice for each $1 \le r \le 12$, and $13/299$ eleven times. For a probability density f, use $p_j = \int_{I_j} f(x)\,\mathrm{d}x$ and $\sum_j p_j = 1$. Equal masses in unequal intervals do not mean equal densities.

2 What the phase measurements can and cannot identify

The classes in (4) partition a phase coordinate. They are not automatically spatial voxels, particle species or independent viewing directions. Such interpretations need additional maps and physical measurement operators.

2.1 Moments require interval widths

For the convention $c_k = \int_0^1 f(x)e^{2\pi ikx}\,\mathrm{d}x$, a piecewise constant reconstruction $f = p_j/w_j$ on I_j gives

$$\widetilde{c}_k = \sum_{j=0}^{34} p_j e^{2\pi ikm_j}\,\mathrm{sinc}(\pi k w_j), \qquad \mathrm{sinc}(u) = \begin{cases} \sin u/u, & u \neq 0, \\ 1, & u = 0. \end{cases} \tag{5}$$

A discrete Fourier transform of the 35 bin indices uses equal index spacing and does not implement this integral on the unequal phase intervals.

Proposition 2.1 (A common invisible mode)**.** *For $0 < \varepsilon < 1$, the distinct probability densities*

$$f_\pm(x) = 1 \pm \varepsilon\cos(2\pi 299x) \tag{6}$$

have identical masses in all 35 classes. Their integrals are also identical in every shifted interval of width $r/299$, for every integer r.

Proof. Every bin boundary is an integer multiple of $1/299$. Integrating the cosine between two such boundaries gives zero. Over any shifted interval of width $r/299$, its antiderivative has equal endpoint values because its phase advances by $2\pi r$. Both densities are nonnegative and integrate to one. □

More observations through an operator with this same nullspace cannot reveal the missing mode. Interpolation of existing bin masses supplies no new independent data.

2.2 A conditional remedy with an additional measurement channel

For a periodic moving-window integral $g_w(s) = \int_s^{s+w} f(x)\,\mathrm{d}x$, the response to $e^{2\pi ikx}$ is

$$h_k(w) = e^{\pi ikw}\frac{\sin(\pi kw)}{\pi k}, \qquad h_0(w) = w. \tag{7}$$

Assume a declared finite band $|k| \le K$. An *actual additional* width $w_* = 1/(2K)$ has, for $1 \le |k| \le K$,

$$|h_k(w_*)| \ge \frac{1}{\pi K}, \tag{8}$$

using $\sin u \ge 2u/\pi$ on $[0, \pi/2]$. Enough independent shifts are still needed to recover the band coefficients. If the resulting finite measurement matrix M has smallest singular value $\sigma_{\min} > 0$, least-squares inversion obeys

$$\left\|\widehat{f} - f\right\| \le \frac{\|\eta\|}{\sigma_{\min}}. \tag{9}$$

At $K = 299$, the added width is $1/598$. These bounds concern the stated finite model, sampling and noise convention. They do not reconstruct arbitrary hidden frequencies. Here the matrix acts on coefficients in an orthonormal Fourier basis; the bound uses Euclidean coefficient and data norms, equivalently the function's $L^2(\mathbb{T})$ norm. The sampling and weighting normalization are part of M and its singular values.

3 Coupled modes and the meaning of a common framework

Let $z \in \mathbb{C}^N$, let W be a constant positive Hermitian matrix, and consider

$$\dot{z} = -\mathrm{i}Lz, \qquad WL = L^\dagger W. \tag{10}$$

Proposition 3.1 (Weighted norm conservation)**.** *Equation* (10) *conserves* $E_W = z^\dagger W z$*. Moreover,* $\psi = W^{1/2}z$ *satisfies* $\mathrm{i}\dot{\psi} = H\psi$ *with the Hermitian matrix* $H = W^{1/2}LW^{-1/2}$*.*

Proof. Differentiate to obtain $\dot{E}_W = \mathrm{i}z^\dagger(L^\dagger W - WL)z = 0$. The stated condition also gives $H^\dagger = H$. □

This is a useful finite-dimensional construction. Interpreting E_W as joules requires a physical state map, dimensions and calibration. It does not follow merely from calling it energy. Nor does finite-dimensional norm conservation imply a positive continuum mass gap, spatial regularity or an arithmetic identity.

For a damping extension $\dot{z} = -\mathrm{i}Lz - \Gamma z$,

$$\dot{E}_W = -z^\dagger(\Gamma^\dagger W + W\Gamma)z. \tag{11}$$

If the matrix in parentheses is positive semidefinite, the norm decreases. A reservoir variable U with the opposite derivative conserves $E_W + U$. Without a reservoir or boundary flux, deleting amplitude is not a complete physical energy balance.

3.1 A precise role for shared methods

A common notation can organize operators, retained information, symmetries, norms, limits and verification. To transfer a result to a target problem, one needs a problem-specific encoding and a proved connection to the original statement.

Proposition 3.2 (Transport of a predicate)**.** *Let* $E : X \to Z$ *be an encoding and let predicates* P *and* $\widetilde{P}$ *satisfy* $P(x) \iff \widetilde{P}(E(x))$ *for all* $x \in X$*. If* $\widetilde{P}$ *holds on* $E(X)$*, then* P *holds on* X*.*

Proof. Apply the assumed equivalence to each $x \in X$. □

This elementary statement supplies a checklist, not a new solution technique by itself. The equivalence must preserve the full quantifiers and mathematical structures. For existence claims, candidate solutions and admissibility must also be transported. For complexity, the encoding, calculation and decoding costs must be bounded. For continuum problems, convergence must preserve the required estimates and axioms.

Required bridge. No established reduction in the source paper turns all six targets into one proved statement about (10). A useful common lemma would have to be stated and proved independently, then connected to each target by six verified bridges.

4 Navier–Stokes: conservation is weaker than regularity

The official target concerns the three-dimensional incompressible equations

$$\partial_t u + (u \cdot \nabla)u = \nu \Delta u - \nabla p + f, \qquad \nabla \cdot u = 0, \quad \nu > 0. \tag{12}$$

The formulation permits smooth global existence for every admissible smooth unforced initial velocity on $\mathbb{R}^3$ or the periodic domain, or an admissible breakdown example in one of those settings, with the specified smooth forcing allowed in the breakdown alternatives [4]. These domain, data, decay and smoothness requirements are part of the problem. A pipe experiment changes the boundary-value setting.

For smooth unforced periodic solutions, integration by parts gives

$$\tfrac{1}{2}\,\|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 \,\mathrm{d}s = \tfrac{1}{2}\,\|u(0)\|_2^2 . \tag{13}$$

The convection and pressure terms have zero net contribution in this setting. Viscosity transfers resolved kinetic energy to internal energy in a fuller thermodynamic model.

Proposition 4.1 (A norm does not control all gradients)**.** *On* $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ *with normalized volume measure, the divergence-free fields*

$$u_k(x) = (0, \sin(kx_1), 0), \qquad k = 1, 2, \ldots,$$

satisfy $\|u_k\|_2 = 1/\sqrt{2}$ *and* $\|\nabla u_k\|_2 = k/\sqrt{2}$.

Proof. The only nonzero velocity component is independent of x_2, so its divergence is zero. The mean squares of sine and cosine over the torus are $1/2$. □

This family is an obstruction to inferring derivative bounds from a conserved zeroth-order norm. It is not a singular Navier–Stokes solution and is not a counterexample to global regularity.

4.1 A possible UNIQUANT approximation

For divergence-free basis fields φ_j, propose

$$u_N(x,t) = \sum_{j=1}^{N} a_j(t)\varphi_j(x). \tag{14}$$

Time addressing may use (A, ρ); phase bins may store selected statistics. A valid analysis must retain nonlinear interactions, pressure constraints, viscosity and the initial-data class. One must then establish consistency and bounds sufficient for the desired limit as $N \to \infty$ and any time discretization is refined.

Required bridge. Construct a map from the proposed states to admissible velocity fields and prove estimates strong enough to obtain the required smooth solutions, uniformly under refinement, or construct a breakdown example satisfying the official conditions. Equation (10) and its finite norm identity do not provide those estimates.

Status in this paper. The paper gives a representation and a counterexample to an insufficient norm argument. It does not evaluate or reproduce the externally announced 2026 proof.

5 Flow in a pipe: where an additional field would enter

The proposal that overlapping external streams change the energy in a pipe is a physical hypothesis separate from the unforced mathematical target. It can be made explicit without assuming its truth.

Let ϱ be mass density, u velocity, e specific internal energy, p pressure, τ_v symmetric viscous stress and q_h heat flux. In a fixed inertial control volume define $\mathcal{E}_f = \varrho(e + |u|^2/2)$. Let b be a known body acceleration. Proposed additional coupling is represented by force density f_c and heating density Q_c:

$$\partial_t \varrho + \nabla \cdot (\varrho u) = 0, \tag{15}$$

$$\partial_t(\varrho u) + \nabla \cdot (\varrho u \otimes u + pI - \tau_v) = \varrho b + f_c, \tag{16}$$

$$\partial_t \mathcal{E}_f + \nabla \cdot \left[(\mathcal{E}_f + p)u - \tau_v u + q_h\right] = \varrho b \cdot u + f_c \cdot u + Q_c. \tag{17}$$

Here f_c has units $\mathrm{N\,m^{-3}}$ and Q_c has units $\mathrm{W\,m^{-3}}$. Moving walls or control volumes require the corresponding boundary-work and transport terms. They cannot be represented by a silent change of coordinates.

If the additional field is part of a closed model, its energy must obey a complementary balance, for example

$$\partial_t \mathcal{E}_c + \nabla \cdot J_c = -f_c \cdot u - Q_c. \tag{18}$$

Adding (17) and (18) cancels internal exchange. A new force also requires a compatible momentum balance for the field or its source.

5.1 Counterflows and unresolved energy

In a spatial cell C, let p_j be *mass fractions*, $\bar{v}_j$ conditional mean velocities and $s_j = \mathbb{E}[|v|^2 \mid j]$. With cell mass M_C,

$$\bar{u} = \sum_j p_j \bar{v}_j, \qquad K_{\mathrm{unres}} = \frac{M_C}{2}\left(\sum_j p_j s_j - |\bar{u}|^2\right) \geq 0. \tag{19}$$

This follows from variance nonnegativity. Equal opposing streams can have zero mean velocity and positive kinetic energy. The unresolved part can include coherent counterflow or turbulent motion; assigning all of it to heat needs an additional physical argument. Phase probabilities from Section 1 become these mass fractions only if an explicit measurement and state map justify that identification.

At a steady junction, continuity requires the outgoing mass flow to equal the sum of incoming mass flows. Redirection and mixing may convert mechanical energy into internal energy and increase entropy. “Two directions into one” alone neither quantifies a pressure loss nor establishes a missing cosmic source term.

6 Material, charge, orientation and frequency

Different objects need not respond equally to the same *physical* field. Their response depends on the interaction, mass, geometry, charge and internal structure. The following constitutive model makes these proposed distinctions explicit.

Hypothesis 6.1 (A candidate superposed contribution)**.** *One may parameterize a proposed electric-field contribution as*

$$E_c(x,t) = \mathrm{Re}\sum_j a_j \epsilon_j e^{\mathrm{i}(k_j\cdot x - \omega_j t + \phi_j)}.$$

The amplitudes have electric-field units, ϵ_j is a dimensionless polarization vector, and a dispersion relation must connect k_j and ω_j. This is a test model, not evidence for a new cosmic field. Electromagnetic interpretation also requires compatible magnetic fields, sources and Maxwell equations. A different type of field needs its own units and coupling law.

6.1 Frequency-selective, anisotropic response

For object a, let $R_a \in SO(3)$ describe its physical orientation. A linear dipole response to an electric field can be written

$$p_a(\omega) = \alpha_a^{\mathrm{lab}}(\omega)E(\omega), \qquad \alpha_a^{\mathrm{lab}}(\omega) = R_a \alpha_a^{\mathrm{body}}(\omega) R_a^T. \tag{20}$$

The orientation is treated as fixed during each linear-response measurement. A time-dependent rotation generally needs a time-dependent response model and can mix frequencies. The polarizability tensor may be complex and frequency dependent. It describes a material response; it is not specified by the UNIQUANT time grid. Polarization of the incident field matters through its direction relative to this tensor. Changing the body orientation can alter absorption, forces or torque. Rotating only the coordinate axes, while transforming all vectors and tensors consistently, cannot change a physical prediction.

For a net charge Q_a, one contribution is the Lorentz force

$$F_a^{\mathrm{charge}} = Q_a(E + v_a \times B). \tag{21}$$

For a fixed electric dipole in a quasistatic applied field, a force contribution is $\nabla(p_a \cdot E)$ and a torque is $p_a \times E$. Induced dipoles require the appropriate response energy and averaging; permanent and induced dipole formulas must not be interchanged. A general measured response kernel is safer outside these approximations.

For a small electric dipole with peak-amplitude convention $E(t) = \mathrm{Re}(E_0 e^{-\mathrm{i}\omega t})$, the mean work extracted from the applied field by its dipole response is

$$\overline{P}_a = \frac{\omega}{2}\,\mathrm{Im}\left(E_0^* \cdot \alpha_a^{\mathrm{lab}}(\omega)E_0\right). \tag{22}$$

Material absorption and reradiation must be separated when converting this work into heat. A passive response cannot be assigned arbitrary negative dissipation. For broadband fields, the response must be integrated with a declared spectral convention; coherent overlapping components can have cross terms.

6.2 Density differences require a specified force law

The relative acceleration of two bodies is

$$\Delta a = F_1/m_1 - F_2/m_2. \tag{23}$$

For example, the same pressure force on equal volumes gives different accelerations when their mass densities differ. Charge coupling scales with charge per mass, not mass density alone. Equation (20) permits different resonances and preferred directions. A claim of universal gravitational acceleration must be distinguished from such material-dependent interactions.

For small displacements about a specified reference trajectory, a separate mechanical response function can be defined by

$$\delta x_a(t) = \int_0^\infty G_a(s) E_c(t-s)\,\mathrm{d}s, \qquad \delta x_a(\omega) = \chi_a(\omega) E_c(\omega). \tag{24}$$

Within a specified linear regime where the response exists, the kernel is causal. The units of χ_a are displacement divided by field strength. It includes the mechanical dynamics as well as field coupling. It is not the dipole polarizability α_a. If the field is approximately uniform across two bodies, then

$$\delta x_1(\omega) - \delta x_2(\omega) = [\chi_1(\omega) - \chi_2(\omega)] E_c(\omega). \tag{25}$$

Equal responses cancel in this differential channel. Different resonances, charges or physical orientations can leave a measurable residual with a predicted amplitude and phase. For spatially varying fields, evaluate the field separately at each body.

Required bridge. Specify the candidate field, its spectrum and polarization, a causal response kernel, coupling strengths, and conserved energy–momentum exchange. Predict the differential signal in (23) before using it to explain an observation.

7 Shared motion and distance-dependent errors

At equal coordinate time, an elementary relative-position model is

$$n = \frac{x - o}{D}, \qquad D = |x - o|, \tag{26}$$

where o is the observer and x the target. A common translation $x \mapsto x + b(t)$, $o \mapsto o + b(t)$ leaves this instantaneous relative direction unchanged. This is a measurement degeneracy; it does not prove that an external force exists. An accelerating coordinate frame also changes the dynamical equations through inertial terms. Actual light-time, aberration and clock models must be handled explicitly.

For small independent position changes,

$$\delta n \simeq \frac{I - nn^T}{D}(\delta x - \delta o). \tag{27}$$

Thus material-dependent response can break a shared-motion degeneracy. Known orientation, field modulation or independent ranging can help distinguish the responses. If every relevant observable remains invariant, the data cannot identify the proposed common mode without an additional measurement channel.

7.1 Two different error scalings

For fixed angular uncertainty σ_θ, the transverse position uncertainty is approximately

$$\sigma_\perp \simeq D\sigma_\theta, \qquad \frac{\sigma_\perp}{D} \simeq \sigma_\theta. \tag{28}$$

Absolute error grows with distance in this model; relative transverse error does not. For parallax angle π and known baseline B, the small-angle relation $D \simeq B/\pi$ instead gives

$$\frac{\sigma_D}{D} \simeq \frac{\sigma_\pi}{|\pi|}. \tag{29}$$

This linear propagation requires errors small compared with $|\pi|$. Near zero parallax, direct inversion is unstable and a full inference model is necessary. There is no single universal "farther means proportionally more wrong" rule.

7.2 An observer-aware inverse problem

Use measurements y_{ib} in observer configuration i and frequency band b:

$$y_{ib} = H_b\big(x(t_i;\theta), o_i(t_i), R_i(t_i), \gamma_b, \beta\big) + \eta_{ib}. \tag{30}$$

Here γ_b contains instrument parameters and β candidate field-coupling parameters. Fix coordinate conventions, then examine the Jacobian and its singular values. Full column rank with respect to all unknown parameters, after fixing coordinate freedoms, is a sufficient regular local identifiability condition, not a guarantee of global uniqueness. Correlated calibration errors belong in the measurement covariance and need not average away across views.

Spacecraft attitude and calibration are already explicit in Gaia processing [10]. Planetary ephemerides use range, Doppler, VLBI and lunar laser measurements, not a single static optical view [11]. A proposed correction must predict independent channels consistently rather than reinterpret only one image.

7.3 A moving observer and a fixed observation point

Let a candidate scalar mode amplitude be

$$\Psi(x,t) = \sum_j a_j e^{i\Phi_j(x,t)}, \qquad \Phi_j(x,t) = k_j \cdot x - \omega_j t + \phi_j. \tag{31}$$

The amplitude sampled along an observer trajectory $x_o(t)$ is its pullback

$$\Psi_{\rm obs}(t) = \Psi(x_o(t),t), \qquad \omega_{j,\rm eff}(t) = -\frac{\mathrm{d}}{\mathrm{d}t}\Phi_j(x_o(t),t) = \omega_j - k_j \cdot \dot{x}_o(t). \tag{32}$$

The exact sampling derivative is $\mathrm{d}\Psi_{\rm obs}/\,\mathrm{d}t = (\partial_t + \dot{x}_o \cdot \nabla)\Psi|_{x_o(t)}$. The displayed frequency is per component. A superposition generally has no unique frequency, and its phase is undefined at interference zeros. For a plane-wave phase in flat spacetime, the proper-time rate is $\omega_{j,\rm obs} = \gamma_o(\omega_j - k_j \cdot v_o)$, where $\gamma_o = (1 - |v_o|^2/c^2)^{-1/2}$. Gravitational clock effects, changing polarization projections and the instrument response require additional terms in the measurement model. Scalar phase is invariant; electromagnetic field components must be transformed as electromagnetic fields.

If $x_o(t) = x_0$, the point is fixed in those coordinates, but $\Psi(x_0,t)$ still contains the temporal superposition. Physically holding a detector fixed requires the appropriate support forces and work balance. If instead

$$x_o(t) = x_0 + \sum_r \xi_r \cos(\Omega_r t + \vartheta_r), \tag{33}$$

the phase contains $\sum_r (k_j \cdot \xi_r)\cos(\Omega_r t + \vartheta_r)$ and is modulated by the observer motion. The amplitudes ξ_r and frequencies Ω_r must follow a specified response or trajectory model; they do not follow just from writing field modes.

For the scalar coordinate change $x' = x - b(t)$, define

$$\Psi'(x',t) = \Psi(x' + b(t), t), \qquad x'_o(t) = x_o(t) - b(t). \tag{34}$$

Then $\Psi'(x'_o(t),t) = \Psi(x_o(t),t)$. Choosing $b = x_o$ fixes the observer at the new origin and transfers the time dependence into the field representation. These are two descriptions of the same sampling, not two independent energy sources. Dynamical equations in an accelerating frame must include their inertial terms.

7.4 Local energy must include the actual superposition

For real electromagnetic fields in vacuum, in one specified local inertial frame,

$$u_{\rm EM} = \frac{\epsilon_0}{2}|E|^2 + \frac{1}{2\mu_0}|B|^2, \qquad S = \frac{1}{\mu_0} E \times B. \tag{35}$$

With $E = \sum_j E_j$ and $B = \sum_j B_j$, the energy contains cross terms:

$$u_{\rm EM} = \sum_j u_j + \epsilon_0 \sum_{j<\ell} E_j \cdot E_\ell + \frac{1}{\mu_0} \sum_{j<\ell} B_j \cdot B_\ell. \tag{36}$$

The scalar amplitude needs a map to physical fields before it predicts energy. Evaluate those fields at the observation event and use the detector's averaging interval. Incoherent cross terms may average away; coherent ones can survive. Local field energy is distinct from energy absorbed by a body, which needs the coupling law. A proposed non-electromagnetic stream needs its own energy, flux and observer transformation.

8 Yang–Mills: from finite modes to a quantum field theory

For every compact simple gauge group G, the official target asks for a nontrivial quantum Yang–Mills theory on $\mathbb{R}^4$ meeting the required quantum-field-theory axioms, with a positive mass gap [5]. In a Hamiltonian description with vacuum energy zero, the spectral requirement is

$$\operatorname{spec}(H) \subset \{0\} \cup [\Delta, \infty), \qquad \Delta > 0. \tag{37}$$

The classical expressions

$$F_A = \mathrm{d}A + A \wedge A, \qquad D_A^* F_A = 0 \tag{38}$$

describe a gauge curvature and a classical field equation. Writing these equations or constructing a finite Hermitian matrix does not construct the required quantum theory.

8.1 A gap can disappear under refinement

Proposition 8.1 (Positive finite gaps need not persist)**.** *On increasing finite-dimensional subspaces of ℓ^2, the Hermitian matrices*

$$H_n = \operatorname{diag}(0, 1, 1/2, \ldots, 1/n), \qquad n \geq 2,$$

generate norm-preserving evolution and each has positive ground-state gap $1/n$. The limiting bounded operator $H = \operatorname{diag}(0, 1, 1/2, 1/3, \ldots)$ has no positive gap above zero.

Proof. The smallest positive eigenvalue of H_n is $1/n$. The limiting diagonal operator is self-adjoint and has positive eigenvalues $1/j$ arbitrarily close to zero, so no positive Δ excludes spectrum in $(0, \Delta)$. Hermiticity gives norm-preserving evolution independently of this gap. Extending H_n by zero on the remaining coordinates gives $\|H - H_n\|_{\mathrm{op}} = 1/(n+1)$. □

This example does not model a gauge field. It isolates the logical failure in inferring a continuum gap merely because every numerical approximation has one.

8.2 Possible work within a common representation

A useful discretization would need gauge-compatible variables, observables and constraints. If it uses a mesh spacing a, volume size V and basis cutoff N, it must specify what physical quantities remain fixed while $a \to 0$, $V \to \infty$ and $N \to \infty$. Changing energy units with resolution can manufacture an apparent dimensionless gap; the bound must survive in fixed physical units.

Required bridge. Construct the nontrivial limiting quantum theory, prove the required axioms and show a strictly positive physical mass gap in the limit. Gauge compatibility, continuum existence, infinite-volume control and the gap are distinct obligations.

Status in this paper. Weighted modes supply finite linear algebra and possible approximation variables. Neither a quantum Yang–Mills construction nor a uniform mass-gap proof is provided here. The proposed classical external-flow hypothesis is also not such a construction.

9 P versus NP: representation costs remain real costs

For input $x \in \{0,1\}^n$, P consists of decision problems solvable by a deterministic algorithm in worst-case time polynomial in n. Membership in NP has a polynomially bounded certificate verified in polynomial time:

$$x \in L \quad \Longleftrightarrow \quad \exists y,\ |y| \leq p(|x|),\ V(x,y) = 1. \tag{39}$$

The target is to prove or disprove $P = NP$ [6]. A new time address, phase code or wave representation is an encoding; it is not yet an algorithm that decides a language with the required complexity.

Proposition 9.1 (An exact code cannot erase arbitrary information)**.** *If every n-bit string is injectively encoded in a fixed b-bit representation, then $b \geq n$.*

Proof. There are 2^n distinct inputs and only 2^b possible codes. Injectivity requires $2^n \leq 2^b$. □

This counting statement does not exclude useful compression of restricted families, variable-length coding or algorithmic structure. It excludes lossless fixed-length compression of *every* input into fewer bits.

For example, one can place an n-bit word in a single rational number

$$r = \frac{\mathrm{int}(x)}{2^n}. \tag{40}$$

Recovering every word still needs the length n and sufficient precision to distinguish adjacent values spaced 2^{-n}. One real coordinate is not one bit. An analog device also needs preparation, readout and noise budgets. Assuming exact arbitrary real arithmetic at unit cost changes the computational model and does not resolve the standard bit-complexity question.

9.1 A valid algorithmic route

For a proposed SAT solver, specify an input encoding, deterministic update rule, termination criterion and decoding rule. Prove correctness for every instance, then bound the number of operations and the bit lengths of every intermediate quantity. The total bound must include addressing, matrix construction, precision, arithmetic, memory access and output verification. Testing many instances can expose a failure; it cannot establish a worst-case asymptotic theorem.

Required bridge. To establish $P = NP$, give a correct deterministic polynomial-time algorithm for an NP-complete problem in the standard model. To establish $P \neq NP$, prove that some language in NP has no deterministic polynomial-time algorithm. Failures of a selected family of methods do not rule out all algorithms.

Status in this paper. UNIQUANT's exact rational coordinates can be implemented and costed. No polynomial SAT algorithm or separating lower bound is derived in this paper.

10 The Riemann Hypothesis: symmetry needs an exact bridge

For $\operatorname{Re} s > 1$, define $\zeta(s) = \sum_{n\geq 1} n^{-s}$, then use its meromorphic continuation. The Riemann Hypothesis states

$$\zeta(\rho) = 0, \quad 0 < \operatorname{Re}\rho < 1 \quad \Longrightarrow \quad \operatorname{Re}\rho = \tfrac{1}{2}. \tag{41}$$

The completed entire function is

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \qquad \xi(s) = \xi(1-s). \tag{42}$$

These are analytic statements about one particular function [7]. A symmetric diagram, finite Fourier fit or finite list of checked zeros does not settle the global zero statement.

Proposition 10.1 (Reflection symmetry does not fix the zero line)**.** *The polynomial*

$$F(s) = \left[(s-\tfrac{1}{4})^2 + 1\right]\left[(s-\tfrac{3}{4})^2 + 1\right] \tag{43}$$

obeys $F(1-s) = F(s)$ *and* $F(\bar{s}) = \overline{F(s)}$*, but its zeros are* $1/4 \pm \mathrm{i}$ *and* $3/4 \pm \mathrm{i}$.

Proof. The substitution $s \mapsto 1-s$ exchanges the two factors. Real coefficients give conjugation symmetry. Solving either quadratic gives the listed roots. □

This is a counterexample to a symmetry-only inference, not a counterexample to the Riemann Hypothesis. The arithmetic and analytic structure of ζ cannot be replaced by generic reflection symmetry.

10.1 A conditional spectral approach

If a suitable self-adjoint operator $\mathcal{H}$ had a proved, globally complete correspondence between its real spectral values and

$$\lambda_\rho = \frac{\rho - 1/2}{\mathrm{i}} \tag{44}$$

for *every* nontrivial zero, the reality of λ_ρ would force $\operatorname{Re}\rho = 1/2$. However, self-adjointness of an arbitrary finite matrix does not establish that correspondence. A proposal must specify the Hilbert space, operator domain and exact analytic identity, including all zeros and multiplicities.

Required bridge. Construct and justify an exact link to ξ, with the required global completeness and analytic control. The weighted Hermitian matrix from (10) has no demonstrated zero-preserving connection to ζ.

Status in this paper. The paper proves the symmetry obstruction and describes the missing spectral bridge. It proves no new statement about the locations of zeta zeros.

11 Hodge: an algebraic cycle is not a periodic signal

For a smooth projective complex variety X, the rational Hodge conjecture asks for surjectivity of the cycle-class map

$$\mathrm{cl} : CH^p(X) \otimes \mathbb{Q} \longrightarrow H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X) \tag{45}$$

for every codimension p [8]. The right side denotes rational cohomology classes whose complexifications have Hodge type (p, p). The unrestricted integral analogue is not the conjecture.

Here an algebraic cycle is a formal combination of algebraic subvarieties. A temporal cycle, sinusoidal oscillation, closed orbit or loop in a rendered figure is a different object. Similar vocabulary supplies no map between them.

11.1 What a valid common representation would retain

Suppose one tries to encode a cohomology class by phase coefficients or finite measurements. The construction must first identify X and its algebraic structure, the relevant rational cohomology group and the Hodge decomposition. It must distinguish rational classes from arbitrary complex vectors. It must then produce algebraic cycles with precisely the required class, rather than merely smooth approximations to differential forms.

Writing α for a target class, the desired witness has the form

$$Z = \sum_{j=1}^{M} q_j Z_j, \quad q_j \in \mathbb{Q}, \quad \operatorname{codim} Z_j = p, \qquad \mathrm{cl}(Z) = \alpha. \tag{46}$$

Approximating α numerically in a norm is weaker than this exact equality and does not certify that the Z_j are algebraic subvarieties. Harmonic representatives also do not by themselves provide algebraic representatives.

11.2 A restricted initial task

A concrete exploratory project could fix an explicitly described variety, compute known cycle classes and verify that a proposed encoding respects their linear relations. Such a calculation is meaningful when its domain and coefficients are declared. It remains a restricted example; extending to all smooth projective varieties requires a new theorem.

Required bridge. Give a construction that preserves rational cohomology, Hodge type and the cycle-class map, and prove that every target class has an algebraic-cycle preimage. The inverse-problem results of Sections 1–3 do not supply this algebraicity result.

Status in this paper. No new surjectivity theorem is established. The common framework can organize exact data and proof obligations, but cannot identify phase cycles with algebraic cycles without an additional construction.

12 Birch–Swinnerton-Dyer: retain the arithmetic

For an elliptic curve $E/\mathbb{Q}$, the group of rational points has the form $E(\mathbb{Q}) \cong \mathbb{Z}^r \oplus E(\mathbb{Q})_{\text{tors}}$. The rank form of the conjecture is

$$r = \operatorname{ord}_{s=1} L(E, s). \tag{47}$$

At a prime p of good reduction, define

$$a_p = p + 1 - \#E(\mathbb{F}_p), \qquad L_p(E, s) = (1 - a_p p^{-s} + p^{1-2s})^{-1}. \tag{48}$$

The point count includes the point at infinity. The full L-function includes the appropriate factors at bad primes. Refined leading-coefficient statements add further arithmetic quantities and hypotheses; a rank identity and the full refinement must be distinguished [9].

12.1 Geometry is not enough

An elliptic curve here is a nonsingular genus-one algebraic curve with a specified rational point, not an ellipse or a planetary orbit. Over $\mathbb{C}$ it has a torus description, but the rational point group and local reductions are essential to (47). A generic periodic phase representation need not retain them.

A prospective encoding would need maps respecting at least

$$\text{rational points and group law} \quad \longleftrightarrow \quad \text{the proposed encoded arithmetic states.} \tag{49}$$

It must also preserve the local factors defining $L(E, s)$ and justify all analytic operations used at $s = 1$. If an operator is proposed, proving that a spectral multiplicity equals r is only one side of the required relation; the connection to the order of vanishing must also be established.

12.2 A restricted initial task

For an explicitly supplied curve, compare exact rational-point computations and provable rank bounds with carefully certified analytic calculations where available. Record unresolved rank intervals rather than replacing them with a decimal estimate. Checking a finite set of curves can test an implementation and reveal a bad encoding. It cannot establish (47) for every elliptic curve over $\mathbb{Q}$.

Required bridge. Prove an exact bridge between the arithmetic rank and the analytic order of vanishing for all curves in the stated class. A common wave model, a conserved norm or an orbital cycle provides neither side of this bridge automatically.

Status in this paper. No new rank equality, leading-coefficient formula or finiteness theorem for the Tate–Shafarevich group is proved here.

13 Superposed astronomical cycles as a separate test tool

A cycle explorer is useful for checking proposed observation periods and beat patterns. It is not a proof method for the six targets. For declared periods P_j, amplitudes A_j and phases ϕ_j, use

$$s_j(t) = A_j \cos\left(2\pi \frac{t - t_0}{P_j} + \phi_j\right), \qquad S(t) = \sum_{j \in \mathcal{J}} s_j(t). \tag{50}$$

A sine display is a phase change by $-\pi/2$. Components can be toggled individually, provided the meaning of each amplitude and time convention is retained.

For two unit-amplitude components with equal initial phase,

$$\cos(\omega_1 t) + \cos(\omega_2 t) = 2\cos\left(\tfrac{\omega_1 - \omega_2}{2} t\right) \cos\left(\tfrac{\omega_1 + \omega_2}{2} t\right). \tag{51}$$

The magnitude-envelope repeat period is

$$T_{\text{beat}} = \frac{1}{|P_1^{-1} - P_2^{-1}|}. \tag{52}$$

Using the published M2 and S2 tidal angular speeds, this gives approximately 14.7653 days [12]. Such a beat is mathematically expected from the input frequencies. It does not establish an additional physical interaction.

The accompanying explorer uses 70 selectable components, including 37 NOAA tidal constituents, terrestrial and lunar periods, approximate planetary mean angles and idealized solar cycles. Planetary mean-angle rates come from JPL's approximate-element table, whose stated short-table interval is 1800–2050 [13]. These mean-angle differences are not exact conjunction times or full planetary trajectories. Solar periods used as fixed sinusoids are idealized, not exact clocks.

13.1 Why a sum of unit sinusoids is not an energy balance

Tidal potential, irradiance, displacement and electric field have different units and couplings. Their numerical unit-amplitude sums are a pattern explorer. A physical energy prediction requires a model such as Sections 5–6, calibrated amplitudes and the appropriate quadratic or nonlinear terms. Selecting phases after seeing a target pattern can manufacture an apparent match.

13.2 A discriminating experiment

Prespecify a field-coupling law, frequency bands and expected material/orientation responses. Measure pressure drop, mass flow, inlet and outlet temperatures, wall heat exchange, vibration, electromagnetic fields and timing with calibrated instruments. Fit known thermal, mechanical and instrumental effects first. Then compare a baseline model with the additional coupling on held-out times, orientations, materials or sites. Predicting a response to deliberate rotation or field modulation is stronger than a correlation with a freely selected astronomical period. Any claimed new source must close both energy and momentum budgets within measured uncertainty.

14 Reproducible diagnostics and their logical limits

The source package contains a deterministic Python programme using exact rational arithmetic for combinatorial identities and numerical linear algebra for illustrative finite systems. No astronomical observations, pipe measurements or unreported fitted data are used in the diagnostic figure.

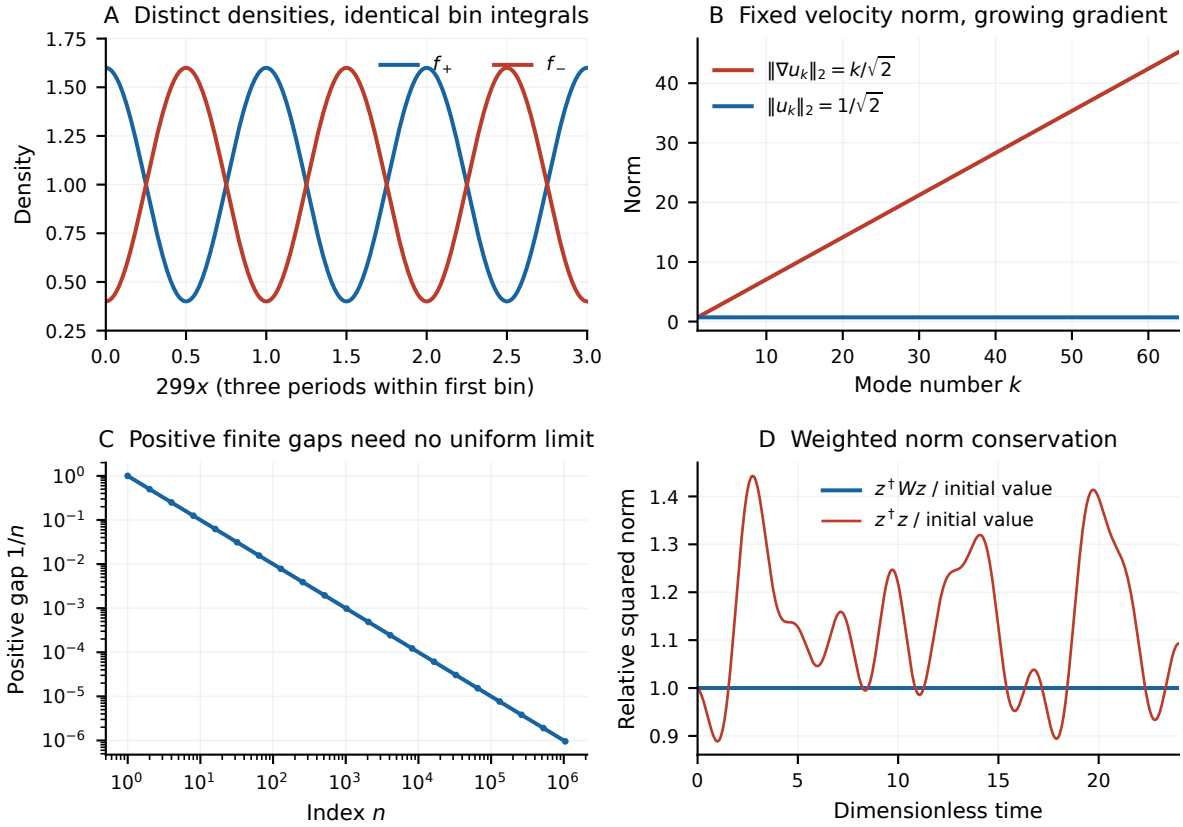


Figure 1: Synthetic diagnostics. Distinct phase densities can have identical bin masses; a bounded velocity norm need not bound gradients; positive finite gaps can shrink to zero; a correctly constructed finite weighted-mode system conserves its norm. These are illustrations of proved statements and limitations, not numerical solutions or experimental tests of the Millennium problems.

Diagnostic	Result	Status
Phase partition	35 bins; widths sum to 1	Exact
Bin null mode	$k = 299$; $\int_{I_j}(f_+ - f_-)\,dx = 0$	Exact
Narrow-window gain	$1/(299\pi) \simeq 0.0010646$	Analytic
Gradient versus velocity	$\|\nabla u_k\|_2/\|u_k\|_2 = k$	Exact
Finite positive gaps	$\Delta_n = 1/n \longrightarrow 0$	Exact
Reflection symmetry	Four off-line zeros of F	Exact
Injective bit encoding	$b \geq n$	Exact
Weighted norm drift	$\max_t \lvert N(t) - N(0)\rvert = 3.6 \times 10^{-15}$	Numerical

The programme also checks the reflection-symmetric polynomial in (43), the exact phase-width counts, selected negative and positive time addresses and the finite weighted-Hermitian construction. Floating-point residuals measure the implementation's numerical error in those examples. They are not bounds for an unconstructed infinite-dimensional limit. The elementary propositions in the text provide the general arguments.

15 A staged programme with explicit stopping rules

The strongest immediate result would be a narrow, independently checkable theorem or a successful prediction of a controlled physical response. It need not settle all six targets to be useful. A shared framework earns its role by preserving the required structure and producing a verifiable new result.

Target	Structure that must survive	Missing decisive result
Navier–Stokes	Divergence constraint, nonlinear PDE, viscosity, admissible data	Required global smooth existence or admissible breakdown; adequate uniform estimates
Yang–Mills	Gauge structure, quantum axioms, physical-unit limits	Nontrivial theory and positive limiting mass gap
P versus NP	Exact bit inputs, correctness and full computation cost	Polynomial NP-complete algorithm or a separating lower bound
Riemann	The particular zeta function and all its nontrivial zeros	Global zero-line theorem or an exact complete spectral bridge
Hodge	Rational cohomology, Hodge type and cycle classes	Algebraic-cycle representatives for every target class
Birch–Swinnerton-Dyer	Rational-point group and arithmetic L-function	Rank equals order of vanishing for every stated curve

Table 1: A common list of obligations does not identify the six structures with each other. Every bridge must be verified in its original mathematical setting.

Stage 1: retain information. Specify state variables, units, exact and approximate quantities, noise models and nullspaces. An identifiability failure must be resolved with a genuinely new channel or an explicitly justified restriction.

Stage 2: test a restricted model. For pipe flow, choose one concrete coupling and independently measurable material response. For pure mathematics, choose one restricted target, retain its original definitions and prove one nontrivial lemma.

Stage 3: control the limit or the cost. State which parameter grows, what remains fixed and which estimates must be uniform. An approximate gap that closes, an unbounded derivative or exponential precision requirement stops the corresponding argument until a new estimate is supplied.

Stage 4: transfer the result. Verify the complete target statement, including its quantifiers, hypotheses and exceptions. A failure on one bridge does not invalidate an unrelated finite-dimensional identity; success on that identity does not repair the failed bridge.

Conclusion. UNIQUANT provides a defined coordinate and measurement framework. The defensible "all at once" objective is coordinated work on information retention, operators and verification. The six mathematical conclusions and the proposed external field effects remain separate obligations. No established common theorem presently connects this construction to all six conclusions.

Attribution, sources and reproducibility

The UNIQUANT research direction, its rational time unit, 13/23 phase construction and proposed physical interpretation are attributed to Joern Christian Anders, as stated in the source working paper. This paper was drafted with AI assistance. Established mathematical definitions and problem statements are cited below; no priority claim is made for standard conservation identities, Fourier methods or linear response theory. The propositions include their assumptions and proofs so that each can be reviewed independently. Scientific and mathematical acceptance requires such review, beyond successful code execution.

The source archive contains `paper.tex`, `reproduce.py`, the diagnostic outputs, an executable build description and a checksum manifest. The supplied script uses Python 3 with NumPy and Matplotlib. Build the diagnostic outputs first, then compile the paper twice with pdfLaTeX. All cited online status information was checked on 6 November 2026; the date matters for the Navier–Stokes announcement.

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